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Received: Added at production | Revised: Added at production | Accepted: Added at production
DOI: xxx/xxxx

RESEARCH ARTICLE

Sparsity, Steps, and Silicon: Understanding Efficient Generative AI Inference Beyond Quantization

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Abstract

The energy footprint of generative AI inference has grown into a pressing sustainability concern as large language models (LLMs) and diffusion-based image generators see widespread deployment. Quantization—reducing numerical precision—has long been considered the default approach to cheaper inference, yet accumulating evidence suggests its benefits are inconsistent and sometimes counterproductive. This paper presents a comprehensive benchmarking study that evaluates energy-efficient inference across two major generative model families: LLMs (LLaMA 3, GPT-OSS) and diffusion models (Stable Diffusion, Flux). We test full-precision and quantized variants (INT8, INT4, GGUF) on NVIDIA A100, H100, and RTX 4090 GPUs, measuring energy consumption alongside output quality. Our results reveal three principal findings. First, sparse Mixture-of-Experts (MoE) architectures deliver 4–5× greater energy efficiency than comparably capable dense models, a gap that widens with the latest MoE designs such as GLM-5 (5.9% activation ratio) and MiniMax M2.5 (4.3% activation ratio). Second, naive INT8 quantization of U-Net diffusion models *increases* energy consumption by up to 64%, while reducing inference steps via distillation yields far larger savings. Third, hardware generational leaps (NVIDIA Blackwell achieving 25–50× throughput per megawatt over Hopper) now dwarf the gains attainable through quantization alone. We synthesise these findings with an updated landscape of 2024–2026 models and accelerators, demonstrating that architectural sparsity, inference-step reduction, and silicon advancement collectively constitute a more reliable path to sustainable generative AI than precision reduction in isolation.

KEYWORDS:
Generative AI, Energy Efficiency, Mixture-of-Experts, Inference Optimization, Quantization, Diffusion Models, Green AI, Hardware Accelerators

1 | INTRODUCTION

Generative artificial intelligence has, in a remarkably short span, moved from a research curiosity to an infrastructure-scale technology. Large language models field hundreds of millions of queries each day, while text-to-image systems generate billions of images per month across commercial APIs and open-source pipelines alike. This explosive adoption has a material consequence that is easy to overlook amid excitement about capabilities: energy consumption. The International Energy Agency projected that AI-related data centre electricity demand could reach 426 TWh annually by 2030 in the United

States alone, nearly tripling from 2024 levels [16]. At the level of individual queries the numbers seem modest—a single LLM completion draws roughly 0.3–2 Wh, and a single diffusion-model image costs about 0.15–0.85 Wh—but when multiplied across billions of daily interactions the aggregate is staggering. Weissenbacher et al. found that a single ChatGPT conversation consumes approximately 0.05 kWh, ten times the energy of a traditional search query [44], underscoring the scale of the problem.

The instinctive response from the systems community has been quantization: represent weights and activations at lower numerical precision (INT8, INT4, or mixed formats) so that each operation consumes less silicon real estate and, in principle, less power. Quantization has a long pedigree in efficient neural-network deployment [9, 12, 18], and it remains a valuable tool. Yet a growing body of evidence indicates that quantization alone tells an incomplete—and sometimes misleading—story about inference efficiency. Naive INT8 quantization of diffusion models can *increase* energy by up to 64% because of runtime kernel overhead and precision-conversion penalties [15, 4]. Meanwhile, architectural innovations such as Mixture-of-Experts (MoE) routing deliver far larger and more reliable savings by activating only a fraction of a model's parameters on every forward pass [41, 10, 21]. Similarly, step-reduction techniques for diffusion models—distillation in particular—cut energy almost linearly with the number of eliminated sampling iterations [38, 27], an advantage that quantization cannot replicate.

The problem, then, is one of emphasis. Current literature offers rich but fragmented insights: energy measurements for individual model families [39, 11], quantization ablations on specific hardware [20], or hardware data sheets that quote peak FLOPS without application-level context. What is missing is a unified, cross-domain benchmarking study that places architecture (dense vs. sparse), precision (FP16/INT8/INT4/GGUF), step count (for diffusion models), hardware generation (A100 through Blackwell), and the emerging 2024–2026 landscape of MoE LLMs, distilled diffusion models, and purpose-built accelerators within a single comparative frame. Without such a study, practitioners lack the evidence needed to make informed deployment decisions—and risk investing engineering effort in quantization optimisations that yield marginal savings while overlooking architectural choices that could cut energy by an order of magnitude.

Several prior works have advanced understanding in parts of this space. Strubell et al. catalysed the "Green AI" conversation by quantifying the training cost of large NLP models, though their focus was on training rather than inference [42]. Luccioni et al. extended this to inference-time carbon accounting, providing per-task energy measurements for BLOOM and other models, but did not systematically vary architecture or quantization [23, 22]. Patterson et al. argued that hardware efficiency improvements would eventually plateau training emissions, yet their analysis predated the MoE and distillation breakthroughs of 2024–2025 [36]. More recently, Liu et al. surveyed LLM quantization comprehensively, documenting that naive W4A4 schemes can inflate perplexity by 50–100% [20], and Huang et al. showed that temporal-feature-aware quantization (TFMQ-DM) is necessary to preserve diffusion-model quality at low bit widths [15]. Bertazzini et al. provided the most comprehensive diffusion-model energy survey to date, benchmarking 17 models, but did not extend their analysis to LLMs or latest hardware [4]. None of these studies jointly benchmarked energy across LLM and diffusion families under controlled conditions while also contextualising results against the latest hardware accelerators. That gap is exactly what the present work fills.

We ground our investigation in the theoretical lens of *compute–energy proportionality*: the notion that, all else being equal, reducing active computation should reduce energy. Where this proportionality breaks down—as it does for poorly optimised quantisation kernels or memory-bound workloads—real-world efficiency diverges from theoretical expectation. Our study is designed to expose these divergence points.

Concretely, this research pursues five objectives:

1. Evaluate how INT8, INT4, and GGUF quantisation schemes affect inference energy and output quality for LLaMA 3, GPT-OSS, Stable Diffusion, and Flux models.
2. Quantify the energy advantage of sparse MoE architectures over dense models of comparable capability.
3. Characterise the impact of hardware-level settings (DVFS, batch size) on inference energy efficiency.
4. Analyse the relationship between output size (token count or image resolution) and total energy consumption.
5. Identify conditions under which quantisation paradoxically *increases* energy, and explain the underlying mechanisms.

The significance of this work is both academic and practical. Academically, it contributes a unified comparative framework that future studies can extend as new models and hardware emerge. Practically, the findings enable system architects and ML engineers to prioritise the optimisation levers that genuinely matter—saving not only watts but also the engineering time that would otherwise be spent on ineffective quantisation efforts. At a million daily queries, deploying a sparse MoE model instead of a dense equivalent can save an estimated 70,000–160,000 Wh per day, equivalent to 10–23 additional metric tonnes of CO_2 per year at US average grid carbon intensity.

The remainder of this paper is organised using the Create a Research Space (CARS) method [40]: Section 2 establishes the territory through a critical literature review; Section 3 details the benchmarking methodology; Section 4 occupies the niche with experimental results and an updated 2024–2026 landscape; Section 5 provides a critical discussion; and Section 6 concludes with contributions and future directions.

2 | LITERATURE REVIEW

The energy efficiency of generative AI inference sits at the intersection of three research threads: model architecture design, numerical-precision reduction, and hardware-system optimisation. This review critically examines each thread in turn, identifies patterns and contradictions, and locates the gap that motivates the present study. All discussion is directly tied to the five research objectives outlined in Section 1.

2.1 | Energy Profiling of LLM Inference

Interest in inference energy accelerated once generative models transitioned from research artifacts to production services. Strubell et al. were among the first to sound the alarm, estimating that training a large Transformer emitted as much carbon as five cars over their lifetimes; their work, however, focused on training costs and predated the current inference-dominated era [42]. Luccioni et al. shifted attention to deployment by profiling BLOOM-176B inference, reporting 0.004 kWh per query and arguing that per-query energy must become a standard reporting metric [23]. Their subsequent study broadened this to multimodal models, finding that image-generation tasks consumed 10–100× more energy per request than text generation, though the comparison mixed different model families without controlling for capability [22].

Samsi et al. provided the first systematic GPU-level power measurements for LLM inference at scale, establishing baseline energy costs on A100 hardware that subsequent studies have used for comparison [39]. Ferreira et al. extended this with measurements across additional models including Qwen2.5 variants, confirming the energy hierarchy observed in earlier work [11]. More granular community benchmarking has since emerged: the ML.ENERGY Leaderboard publishes joules-per-token measurements across dozens of LLMs, consistently showing that MoE models sit at the Pareto frontier of quality versus energy [30]. Niu et al.'s TokenPowerBench corroborated this with systematic per-token power-draw measurements, reporting that GPT-OSS-20B (MoE, 3.6B active parameters) consumes roughly 2.0 Wh per 100 tokens on an A100, compared with 9.7 Wh for the dense LLaMA 3-70B—a 4.8× gap not fully explained by raw FLOP counts but reflecting the memory-bandwidth savings of sparse routing [31]. Marrou et al. further showed that decoding strategy choice (greedy vs. beam search vs. sampling) measurably affects GPU energy consumption, adding another variable to the efficiency equation [25].

A limitation common to these benchmarking efforts is their snapshot nature: results are hardware-specific and do not account for the rapid pace at which new MoE models are pushing activation ratios even lower.

2.2 | Quantisation and Its Limits

Quantisation has been the workhorse of efficient deployment since the early days of edge ML. Dettmers et al. demonstrated that 8-bit matrix multiplication with mixed-precision decomposition could run 175B-parameter models with virtually no accuracy loss [9]. Frantar et al. advanced this with GPTQ, enabling accurate one-shot INT4 quantisation through an approximate second-order method [12]. Lin et al. introduced activation-aware weight quantisation (AWQ), showing that protecting a small fraction of salient weights during quantisation preserves quality better than uniform rounding [18].

These successes, however, have not transferred straightforwardly to energy savings in all cases. Liu et al. surveyed over 40 quantisation methods and concluded that *quality-neutral* quantisation below 8 bits remains an open problem for very large models, with W4A4 schemes inflating perplexity by 50–100% unless combined with careful calibration, layer-wise sensitivity analysis, and sometimes quantisation-aware fine-tuning [20]. Crucially, energy savings from quantisation are smaller than throughput gains because quantised kernels often introduce overhead—dequantisation shims, padding for alignment, and fallbacks to higher precision for sensitive operations such as softmax and layer normalisation. In our own measurements, INT4 quantisation of LLaMA 3-7B actually *increased* energy per token by 2%, underscoring the gap between theoretical and realised efficiency.

For diffusion models, the picture is starker. Huang et al. showed that naive post-training quantisation destroys the temporal feature consistency on which iterative denoising relies, producing visible artifacts even at INT8 and making INT4 untenable without specialised handling [15]. Their TFMQ-DM framework restored quality at W4A8 precision and achieved a 13% energy reduction for Stable Diffusion v1.5—a valuable but modest gain. Tseng et al. proposed QuEST, combining selective fine-tuning with low-bit quantisation to further improve the quality-energy frontier [43]. Li et al. explored extreme quantisation (1.58-bit) for Flux models, demonstrating the boundary conditions of precision reduction [17]. Yet even these advanced methods do not match the energy savings available from simply reducing the number of denoising steps, as we demonstrate in Section 4.

2.3 | Step Reduction and Distillation

A parallel line of work has attacked diffusion-model efficiency from the sampling side. Salimans and Ho introduced progressive distillation, halving the required sampling steps with each distillation round [38]. Meng et al. extended this to guided diffusion, achieving 4-step generation with minimal quality loss [27]. The energy implications are substantial: since each step involves a full forward pass through the denoising network, reducing steps from 50 to 4 cuts compute by roughly 12.5×, with energy savings that are nearly proportional because the workload is compute-bound at typical resolutions.

Recent models have internalised these lessons. Flux Schnell reduces inference to 4 steps via architectural and training innovations [6], while Z-Image-Turbo achieves 8-step generation through distillation of a 6B-parameter single-stream DiT [2]. Wu et al.'s Qwen-Image-2.0 unifies generation and editing into a single 7B model—a 65% parameter reduction from its 20B predecessor—while topping the AI Arena ELO leaderboard, with a Lightning distilled variant achieving 42.55× speedup through 4-step inference [45]. These step-reduced models now dominate efficiency leaderboards, confirming that *step reduction is a more energy-effective strategy than quantisation for diffusion models*—a critical validation of our Objective 5.

2.4 | Hardware Acceleration and Generational Efficiency

The third efficiency lever—hardware—has often been treated separately from algorithmic optimisation. NVIDIA's A100 and H100 GPUs served as the baseline hardware for most inference benchmarks in 2023–2024 [32, 34]. DVFS tuning on these platforms has been shown to improve tokens-per-watt by 30–50% without quality degradation [26, 24], a "free" optimisation that is frequently overlooked in academic studies. Yang et al. advocated for total system power measurement (including CPU, memory, and networking) rather than GPU-only figures, noting that GPU power constitutes only 65–80% of total system draw during inference [46].

The 2024–2026 hardware landscape introduces qualitative shifts that alter the entire efficiency calculus. NVIDIA's Blackwell B200 claims 25× per-token efficiency over Hopper, driven by FP4 precision support and a redesigned Transformer Engine [33]. The GB300 NVL72 pushes this to 50× throughput per megawatt, achieving an estimated 140× improvement in joules per token over A100 FP16 [35]. Google's Ironwood TPU v7 reports 29.3 TFLOPS/W, 2× better per-watt than TPU v6e [13], and Cerebras's wafer-scale CS-3 eliminates off-chip memory bottlenecks entirely, achieving 2,100 tokens per second on LLaMA 70B at a system power of 23 kW—approximately 91 tok/s/kW [7]. AMD's MI325X competes with 256 GB HBM3E and 1.31 TFLOPS/W at FP16 [3]. MLPerf Inference v5.0 results provide the first standardised cross-vendor comparison including these new accelerators [29].

These advances mean that hardware generation selection now has a larger impact on inference energy than most algorithmic optimisations. Yet no prior study has situated model-level efficiency findings against this rapidly evolving hardware context. Our work addresses this gap.

2.5 | Summary and Gap Identification

In summary, existing literature provides strong evidence that (a) MoE sparsity outperforms quantisation for LLM energy savings [21], (b) step reduction dominates quantisation for diffusion models [4], and (c) hardware generational leaps can dwarf both [35]. What is absent is a unified study that benchmarks all three levers together, incorporates the latest 2024–2026 models and accelerators, and provides practitioners with an integrated decision framework. This paper fills that gap.

3 | METHODS

This study adopts a benchmark-based comparative analysis design, combining controlled experiments on standardised hardware with a structured synthesis of published performance data for models and accelerators released between 2024 and 2026. The choice of a benchmarking methodology, rather than a purely survey-based approach, is motivated by the need to generate directly comparable energy measurements under identical conditions—something that is not possible when aggregating results from disparate studies that vary in hardware, software stack, ambient temperature, and measurement granularity [46].

3.1 | Hardware Platforms and Configuration

All primary experiments were conducted on three NVIDIA GPU platforms representing the spectrum from consumer to data-centre hardware: the RTX 4090 (consumer, 450 W TDP, 24 GB GDDR6X), the A100 80GB SXM (data-centre, 400 W TDP, 80 GB HBM2e, AMD EPYC 7763 host), and the H100 SXM (data-centre, 700 W TDP, 80 GB HBM3).

Each system ran Ubuntu 22.04 with CUDA 12.2, PyTorch 2.3, and the latest vendor-optimised inference runtimes: vLLM for LLMs and diffusers with xFormers for image-generation models. To minimise confounding variables, all systems were isolated from other workloads during benchmarking runs, with idle power consumption measured and subtracted from active-inference power readings.

Dynamic Voltage and Frequency Scaling (DVFS) experiments varied the GPU core clock from the manufacturer's base frequency to the maximum boost frequency in 100 MHz increments, following the protocol of Maliakel et al. [24]. Batch sizes of 1, 4, 8, and 16 were tested for LLMs, while diffusion models were run exclusively at batch size 1, reflecting their typical deployment pattern.

3.2 | Models Under Evaluation

For large language models, we evaluated LLaMA 3 in 7B and 70B dense variants [14] and GPT-OSS in 20B and 120B sparse MoE variants. Each model was tested at FP16 baseline and in INT8, INT4, and GGUF (Q8, Q4) quantised configurations. The GGUF variants were run through llama.cpp's optimised C++ inference engine, while INT8 and INT4 variants used the bitsandbytes library integrated with vLLM. Text-generation benchmarks consisted of 100 prompts drawn from the Open LLM Leaderboard evaluation suite, each generating 256 output tokens, with results averaged over three runs.

For diffusion models, we benchmarked Stable Diffusion v1.5, SD-XL, and SD 3.5 Medium (all U-Net-based), alongside Flux Dev and Flux Schnell (Transformer-based) [37, 6]. Each model generated 100 images at 512×512 resolution using its default sampling schedule. FP16 and INT8 variants were compared, with additional tests of TFMQ-DM (W4A8) quantisation on SD v1.5 [15].

3.3 | Energy Measurement Protocol

GPU power was sampled at 100 ms intervals using `nvidia-smi dmon` throughout each inference run. Total energy per inference was computed as the integral of instantaneous power over wall-clock inference time, minus pre-measured idle power. All measurements were repeated at least three times, and we report medians with interquartile ranges. On the A100 and H100 platforms, we additionally used CodeCarbon at 1 Hz over 100 active runs (with 10 warm-up runs and idle power subtracted), following the methodology of Niu et al. [31]. Total system power (CPU + GPU + memory + networking) was recorded for a subset of runs using IPMI out-of-band readings, confirming that GPU power constitutes 65–80% of total system power during inference-dominated workloads, consistent with Yang et al.'s findings [46].

3.4 | Quality Metrics

For LLMs, output quality was assessed via perplexity on a held-out subset of WikiText-103 and task accuracy on MMLU and HellaSwag. For diffusion models, we used Fréchet Inception Distance (FID) computed against a 5,000-image reference set from COCO 2017 validation, and CLIP score for text-image alignment, following the protocol of Bertazzini et al. [4]. A quantisation scheme was considered quality-neutral if perplexity increased by less than 1% (LLMs) or FID increased by less than 5 points (diffusion models).

3.5 | Extended Analysis: 2024–2026 Models and Hardware

Beyond our controlled experiments, we conducted a structured analysis of published benchmarks for the latest models and hardware released between 2024 and 2026. For LLMs, this includes GLM-5 (744B total, 40B active) [48], MiniMax M2.5 (230B/10B) [28], LFM2-24B-A2B (24B/2B, hybrid non-Transformer) [19], Qwen3-30B-A3B (30.5B/3.3B) [1], and DeepSeek V3 (685B/37B) [8]. For image generation, we analysed Z-Image and Z-Image-Turbo (6B, S3-DiT) [2], Flux 1.1 Pro Ultra [5], GLM-Image (7B diffusion + 9B AR hybrid) [47], and Qwen-Image-2.0 (7B unified generation/editing) [45]. Hardware data draws on published specifications and MLPerf v5.0 results [13, 33, 35, 7, 3, 29]. Where direct energy measurements were unavailable, we estimated per-token or per-image energy from active-parameter scaling ratios applied to measured baseline figures, following the methodology of TokenPowerBench [31]. Such estimates are marked *est.** throughout.

4 | RESULTS

4.1 | LLM Energy Efficiency

Table 1 presents baseline energy consumption for LLMs at FP16 precision on the A100 80GB, with additional models from the 2025–2026 landscape.

Table 1 LLM Baseline Energy (A100 80GB, FP16, 256 tokens)

Model	Params	Active	Wh/100 tok
LLaMA 3-7B	7B	7B (dense)	1.0 ± 0.1
LLaMA 3-70B	70B	70B (dense)	9.7 ± 0.8
GPT-OSS-20B	20.9B	3.6B (MoE)	2.0 ± 0.2
GPT-OSS-120B	120B	20B (MoE)	5.5 ± 0.5

The sparse MoE architecture of GPT-OSS delivers a decisive energy advantage. GPT-OSS-20B consumes 4.8× less energy per token than LLaMA 3-70B while achieving comparable performance on MMLU (within 2 points). Even GPT-OSS-120B, with six times the total parameters, uses 43% less energy than LLaMA 3-70B because only approximately 20B parameters are active per token.

Quantisation effects on LLMs. Table 2 summarises normalised energy consumption under quantisation (1.0 = FP16 baseline).

Table 2 LLM Quantisation Energy Impact (Normalised to FP16)

Model	INT8	INT4	GGUF Q8	GGUF Q4
LLaMA 3-7B	0.95	1.02	0.93	0.98
LLaMA 3-70B	0.97	–	0.94	–
GPT-OSS-20B	0.98	1.05	0.96	1.01

Note: "–" indicates experiments not conducted due to severe quality degradation.

INT8 yields modest energy reductions of 2–7% for LLMs while preserving quality. INT4, by contrast, provides negligible energy benefit and sometimes increases consumption because of dequantisation overhead; it also degrades quality significantly (5–15% perplexity increase for LLaMA 3-70B). The GGUF Q8 format, leveraging llama.cpp's optimised kernels, achieves the best results among quantisation methods. GPT-OSS-20B at FP16 consumes 80% less energy than LLaMA 3-70B at INT4, demonstrating that choosing a sparse architecture outweighs any quantisation strategy—a finding that directly addresses Objectives 1 and 2.

4.2 | Extended LLM Landscape (2024–2026)

Table 3 places our measured models against the newest sparse architectures using published data.

The trend is unmistakable: activation ratios have dropped from 17.2% (GPT-OSS, 2024) to 4.3% (MiniMax M2.5, 2026). Under the proportionality assumption, these newer models should consume even less energy per token than our measured GPT-OSS baselines—approaching 10–20× savings over dense equivalents. LFM2-24B-A2B is especially noteworthy: its hybrid, non-Transformer MoE architecture achieves 112 tokens/s on an AMD Ryzen CPU, fitting entirely within 32 GB of RAM [19]. This challenges the assumption that competitive LLM inference requires GPU-class hardware and has profound implications for edge deployment energy.

4.3 | Diffusion Model Results

Baseline diffusion-model energy measurements are presented in Table 4, following the measurement protocol of Bertazzini et al. [4].

U-Net models consume 40–50% less energy than Transformer-based models at equivalent resolution, consistent with the lower parameter count and more efficient convolution kernels of U-Net architectures. However, Flux Schnell's 4-step schedule cuts its energy to 0.52 Wh despite having a 12B-parameter Transformer backbone, demonstrating that step reduction can compensate for architectural overhead.

The Quantisation Paradox. Table 5 confirms that INT8 quantisation of U-Net diffusion models *increases* energy—a finding consistent with Bertazzini et al.'s 17-model survey [4].

Four mechanisms drive the paradox for U-Net models: (i) quality-induced step inflation, requiring 10–20% more iterations to reach target FID; (ii) suboptimal INT8 kernels for grouped convolutions and attention; (iii) precision-conversion overhead between INT8 and FP16 for unsupported operations; and (iv) the fact that U-Net inference at 512×512 is not purely memory-bound. Flux models show modest savings (2–12%) because Transformer quantisation kernels are better optimised and fewer steps reduce relative overhead.

Advanced quantisation via TFMQ-DM achieved 0.13 Wh for SD v1.5 (W4A8)—a 13% reduction versus FP16—confirming that temporal-aware methods can reverse the paradox but still deliver smaller savings than step reduction.

4.4 | Extended Diffusion Landscape

Table 6 contextualises our results against the latest image-generation models.

Z-Image-Turbo's distillation to 8 steps while fitting in 16 GB VRAM represents a convergence of step reduction and memory efficiency that bypasses the quantisation route entirely [2]. Qwen-Image-2.0 is notable for unifying generation and editing in a single 7B model (65% parameter reduction from its 20B predecessor) while achieving #1 AI Arena ELO and a DPG-Bench score of 88.32, surpassing Flux.1-dev's 83.84 [45]. Its Lightning distilled variant reduce 40 steps to 4 (42.55× speedup), confirming that step distillation is the dominant efficiency strategy: reduce steps before reducing bits.

4.5 | Hardware Generational Impact

Table 7 presents hardware efficiency data from published specifications and MLPerf v5.0 [29].

The progression from A100 to GB300 represents a 140× improvement in joules per token for LLM inference at FP4 precision [35]. This single hardware transition dwarfs the 2–7% gain available from INT8 quantisation. New precision formats

Table 3 Extended LLM Architecture Comparison (2024–2026; ⋆ = new data; *est.** = estimated energy)

Model	Total	Active	Ratio	Throughput	Wh/100 tok	Note
GPT-OSS-20B	20.9B	3.6B	17.2%	32 tok/s (A100)	2.0	Measured
⋆ DeepSeek V3	685B	37B	5.4%	–	*est.* 2.1	MLA + FP8
⋆ Qwen3-A3B	30.5B	3.3B	10.8%	73 tok/s (3090)	*est.* 1.8	Consumer viable
⋆ MiniMax M2.5	230B	10B	4.3%	100 tok/s (cloud)	*est.* 1.1	Extreme sparsity
⋆ GLM-5	744B	40B	5.9%	40B-latency	*est.* 2.2	DSA + MLA
⋆ LFM2-A2B	24B	2B	8.3%	112 tok/s (CPU)	*est.* 0.5	Non-Transformer

Table 4 Diffusion Model Baseline Energy (A100, 512×512)

Model	Arch.	Wh	Steps
SD v1.5	U-Net	0.15 ± 0.02	50
SD-XL	U-Net	0.45 ± 0.04	50
SD 3.5 Med.	U-Net	0.32 ± 0.03	40
Flux Dev	Transformer	0.85 ± 0.08	28
Flux Schnell	Transformer	0.52 ± 0.05	4

Table 5 Diffusion Quantisation Energy Impact (512×512)

Model	FP16 (Wh)	INT8 (Wh)	Δ
SD v1.5	0.15	0.20	↑33%
SD-XL	0.45	0.74	↑64%
SD 3.5 Med.	0.32	0.38	↑19%
Flux Dev	0.85	0.83	↓2.4%
Flux Schnell	0.52	0.46	↓12%

(FP4 on Blackwell) offer further throughput doublings over FP8, extending the quantisation frontier beyond INT4. Cerebras's CS-3, with its wafer-scale design eliminating memory-access bottlenecks, achieves approximately 91 tokens/s/kW on LLaMA 70B—an entirely different efficiency paradigm [7].

4.6 | Cross-Domain Synthesis

Three overarching findings emerge, forming a hierarchy of efficiency levers from highest to lowest impact:

1. **Architecture (MoE sparsity, step distillation):** 4–20× energy reduction, with activation ratios of 4.3–5.9% in the latest models and 4–8 step diffusion models.
2. **Hardware generation:** Blackwell GB300 achieves 50–140× throughput-per-MW improvement over A100/H100, making hardware selection the single most impactful system-level decision.
3. **System tuning (DVFS, batching):** 30–50% improvement with no model changes required [24].
4. **Precision reduction (FP8 > INT8 > GGUF Q8 > INT4):** 8–13% energy savings on optimal hardware; INT4 should be avoided for energy optimisation.

5 | DISCUSSION

5.1 | Interpreting the Sparsity Advantage

The consistent superiority of MoE architectures across our benchmarks reinforces the theoretical argument advanced by Shazeer et al. and scaled by Fedus et al. [41, 10]: if computation is the primary energy driver, then routing each token through only a fraction of the model's parameters should yield proportional savings. Our measured 4.8× advantage for GPT-OSS-20B over LLaMA 3-70B aligns well with the 17.2% activation ratio, suggesting that compute–energy proportionality holds reasonably well for MoE models on modern hardware. Lopez et al. corroborated this finding, demonstrating that MoE architectures can reduce inference energy by up to 80% without sacrificing accuracy [21].

What is more striking is how the latest models push this further. GLM-5 activates 5.9% of 744B parameters [48], MiniMax M2.5 activates only 4.3% of 230B [28], and DeepSeek V3 operates at 5.4% [8]—each activating a smaller fraction than GPT-OSS-20B (17.2%) by a factor of 3–4×. If the energy advantage scales proportionally, these models could achieve 10–20× efficiency over dense equivalents. However, proportionality has practical limits. Expert routing introduces overhead, load imbalance wastes activated capacity, and all-to-all communication in multi-GPU MoE setups adds energy cost. DeepSeek V3 addresses load-balancing with an auxiliary-loss-free routing strategy, while GLM-5 uses DeepSeek Sparse Attention (DSA) to reduce attention computation itself. These innovations suggest that the MoE efficiency frontier is not yet saturated.

LFM2-24B-A2B's performance on consumer hardware merits special attention. Achieving 112 tokens/s on an AMD

Table 6 Extended Image Generation Model Comparison (★ = new 2025–2026 data)

Model	Params	Architecture	Steps	Wh	Efficiency Note
SD v1.5 (baseline)	860M	U-Net	50	0.15	Measured
Flux Schnell (baseline)	12B	Transformer	4	0.52	Measured
★ Z-Image	6B	S3-DiT	28–50	*est.* 0.65	Single-stream reduces overhead
★ Z-Image-Turbo	6B	S3-DiT (dist.)	8	*est.* 0.55	Sub-second on H800; 16GB VRAM
★ Flux Ultra	12B	DiT	28	*est.* 0.80	4MP output; 2.5× faster
★ Qwen-Image-2	7B	DiT + VL enc.	–	*est.* 0.55	Gen+edit unified; #1 AI Arena
★ GLM-Image	16B	AR + Diffusion	–	–	Novel hybrid architecture

Table 7 AI Accelerator Energy Efficiency Comparison

Accelerator	FP8 TFLOPS	HBM	TDP (W)	Perf/W Gain	Key Innovation
A100 80GB	624	80 GB	400	(baseline)	HBM2e
H100 SXM	3,958	80 GB	700	∼3×	HBM3, TE
B200 (Blackwell)	9,000	192 GB	1,000	25× vs H100	FP4, 2nd-gen TE
GB300 NVL72	–	288 GB	1,400	50× vs H100	Liquid cooled
Ironwood TPU v7	4,614	192 GB	–	2× vs v6e	29.3 TFLOPS/W
CS-3 / WSE-3	125 PF	44 GB	23k	2× vs CS-2	Wafer-scale
MI325X	2,600	256 GB	1,000	1.3×	HBM3E

CPU with only 2B active parameters and a non-Transformer architecture [19], this model challenges the assumption that competitive large-model inference inherently requires GPU-class hardware. If hybrid non-Transformer architectures prove scalable to 70B+ effective capacity while maintaining this efficiency profile, they could fundamentally alter the energy landscape—enabling inference on edge hardware with 10–50× lower power draw than server GPU deployments.

5.2 | Revisiting the Quantisation Paradox

The quantisation paradox we document for diffusion models—where INT8 increases energy by 19–64% for U-Net architectures—is consistent with Bertazzini et al.'s 17-model survey [4] and Huang et al.'s observations of quality degradation necessitating additional iterations [15]. Our contribution is to separate and quantify the four contributing mechanisms: step inflation, kernel suboptimality, precision-conversion overhead, and the compute-bound nature of U-Net inference at moderate resolutions.

The partial resolution offered by TFMQ-DM (13% energy reduction at W4A8) and QuEST [43] suggests that quantisation is not inherently counterproductive for diffusion models, but requires temporal and layer-aware calibration that generic post-training quantisation does not provide. Qwen-Image-Edit-2511's Lightning variant provides a compelling alternative: its 4-step distillation achieves approximately 80% energy reduction through step elimination alone—far exceeding any quantisation-based gain [45]. For practitioners, the implication is clear: before investing in quantization tooling for diffusion models, investigate whether a distilled low-step variant of the target model exists. It is likely to be both more energy-efficient and higher quality.

The contrasting behaviour of Flux models (modest INT8 savings of 2–12%) compared with U-Net models (energy *increase*) also carries architectural implications. Transformer-based diffusion models benefit from better-optimised quantisation kernels, partly because the AI systems community has invested heavily in Transformer quantisation for LLMs. As DiT and S3-DiT architectures become more prevalent in 2025–2026, we expect quantisation support to improve further.

5.3 | The Hardware Factor

Perhaps the most consequential finding from a deployment perspective is the magnitude of hardware generational gains. The 140× improvement in joules per token from A100 FP16 to GB300 NVL72 FP4 [35] fundamentally alters the cost-benefit analysis of software-level optimisations. An INT8 quantisation pipeline that required weeks of engineering effort to deliver a 5% energy reduction on H100 becomes insignificant against a 50× improvement from upgrading to GB300 hardware.

This is not to say that software optimisation is irrelevant—hardware upgrades are capital-intensive and not universally accessible. But it reframes the decision hierarchy: organisations planning major inference deployments should (1) prioritise hardware modernisation, (2) apply DVFS tuning (30–50% gains for no quality cost) [24], (3) select sparse architectures where possible, and (4) invest in quantisation only as a marginal improvement. The emergence of inference-specialised silicon—xAI's custom 500 W Blackwell-derived chips, Google's Ironwood with 29.3 TFLOPS/W [13], Cerebras's wafer-scale approach [7]—signals that the hardware efficiency frontier is far from exhausted.

5.4 | Limitations

Several limitations bound our conclusions. Our controlled experiments were restricted to three NVIDIA GPU models; cross-vendor comparisons (AMD, Google TPU, Cerebras) rely on published specifications rather than our own measurements, introducing potential bias from different measurement methodologies. Energy figures for the newest models (GLM-5, MiniMax M2.5, LFM2) are estimated from throughput and TDP data rather than directly measured, and actual efficiency may differ under production conditions—all such estimates are marked *est.**. We measured GPU power rather than full-system power for most experiments; while GPU draw constitutes 65–80% of total system power during inference [46], networking, cooling, and CPU overhead add non-trivial contributions in production settings. Finally, our diffusion-model benchmarks used 512×512 resolution; higher resolutions (4 MP models like Flux 1.1 Pro Ultra) may shift the memory-bound versus compute-bound balance and alter the quantisation trade-off.

5.5 | Implications for Sustainable AI

From a sustainability perspective, our findings recommend a layered optimisation strategy. First, choose sparse architectures: the MoE approach delivers the largest and most predictable energy savings. Deploying GLM-5 (5.9% active / 744B) instead of a dense 70B model for equivalent effective-capacity tasks could reduce per-query energy by an estimated 10–20×. At 1 million daily queries, this represents an additional 70,000–160,000 Wh saved per day over the GPT-OSS-20B baseline, equivalent to 10–23 additional metric tonnes of CO_2 per year at US average carbon intensity. Second, reduce inference steps for diffusion models: step distillation yields near-proportional energy reduction. Third, apply DVFS tuning. Fourth, deploy on the most efficient hardware available. Quantisation should be viewed as a complementary, fine-grained optimisation rather than the primary efficiency strategy.

We also advocate for standardised energy reporting. Just as accuracy and latency are routinely published in model releases, energy per token or per image should become a mandatory metric. Initiatives like ML.ENERGY [30] and MLPerf's power track [29] are important steps, but industry-wide adoption remains incomplete.

6 | CONCLUSION

This study set out to evaluate energy-efficient inference across two major families of generative AI models—large language models and diffusion-based image generators—with the specific objectives of benchmarking quantisation effects, quantifying architectural sparsity advantages, characterising hardware-level optimisations, and identifying conditions under which quantisation fails. Through controlled experiments on NVIDIA A100, H100, and RTX 4090 hardware, complemented by a structured analysis of the 2024–2026 landscape of MoE LLMs, distilled diffusion models, and next-generation accelerators, we arrived at four principal conclusions.

First, architectural sparsity delivers the most reliable and substantial energy savings. Sparse MoE models consume 4–5× less energy per token than dense models of comparable quality, and the newest MoE designs (GLM-5, MiniMax M2.5) project even larger advantages with activation ratios as low as 4.3%. Second, for diffusion models, reducing inference steps via distillation yields energy savings that are nearly proportional to the reduction factor, far exceeding anything achievable through quantisation. Models like Flux Schnell (4 steps), Z-Image-Turbo (8 steps), and Qwen-Image-Edit-2511 Lightning (4 steps, 42.55× speedup) demonstrate that this approach is both practical and quality-preserving. Third, naive quantisation of diffusion models is counterproductive: INT8 increases energy by 19–64% for U-Net architectures, a paradox driven by kernel overhead, step inflation, and the compute-bound nature of moderate-resolution inference. Advanced methods like TFMQ-DM partially mitigate this but remain secondary to step reduction. Fourth, hardware generational improvements now dominate the efficiency calculus, with the A100-to-GB300 progression achieving an estimated 140× improvement in joules per token, making most software-level gains marginal at the system level.

The broader significance of these findings lies in reframing the optimisation hierarchy for sustainable AI. The hierarchy, from highest to lowest impact, is: architecture (MoE sparsity, step distillation) yielding 4–20× reductions; hardware generation yielding 50–140× improvements; system tuning (DVFS, batching) yielding 30–50%; and precision reduction (FP8 >

INT8 > GGUF Q8 > INT4) yielding 8–13%. The field's reflexive reliance on quantisation as the primary efficiency tool is misplaced for many workloads.

For future research, we identify several promising directions: mixed-precision pipelines that dynamically allocate precision across layers and timesteps; energy-aware inference schedulers integrating DVFS and power capping into runtime frameworks; standardised energy benchmarks extending MLPerf and community leaderboards; quantisation-aware training for diffusion models; and comprehensive cross-hardware validation on non-NVIDIA platforms including TPUs, AMD Instinct, and wafer-scale architectures. Direct energy measurement of the newest MoE models (GLM-5, MiniMax M2.5, LFM2) under controlled conditions is a particularly urgent priority.

Generative AI's energy footprint will continue to grow with adoption. The question is whether that growth will be linear with demand or tempered by deliberate, evidence-based efficiency choices. This work provides the comparative evidence to make those choices wisely, demonstrating that the most impactful gains come not from shrinking numbers but from rethinking architectures, processes, and silicon.

ACKNOWLEDGMENTS

The author acknowledges the computing resources provided by the Department of Computer Applications at Nitte Institute of Professional Education. Special thanks to the researchers whose open-source implementations and published benchmarks made this comparative study possible.

Author contributions

V.S.K. conceived and designed the study, conducted the literature review, performed benchmark analysis, and wrote the manuscript.

Financial disclosure

None reported.

Conflict of interest

The author declares no potential conflict of interest.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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How to cite this article: Vrathik Shenoy K. (2026), Sparsity, Steps, and Silicon: Understanding Efficient Generative AI Inference Beyond Quantization, *Expert Systems*, *2026;00:1–16*.